

WayPoint — Integrated Transit Operations Platform

SE3110: Quality Management in Software Engineering

Software Testing & Quality Evaluation Report

Comprehensive Group and Individual Testing Evaluation of the SE3090 Integrated System

Document Control & Metadata

Submission Attribute	Official Academic & Technical Information
Academic Module	SE3110: Quality Management in Software Engineering
Associated Module	SE3090: Integrated Full-Stack and Agentic AI Application Development
Academic Year / Semester	Year 3, Semester 1 (Academic Year 2026)
Institution	Faculty of Computing — Sri Lanka Institute of Information Technology (SLIIT)
Project Title	WayPoint (Intercity Transit Operations & Journey Planning Platform)
Document Purpose	Comprehensive Software Testing, Quality Evaluation & Defect Remediation Report
Date of Submission	8th October 2026
Total Automated Tests	448 Verified Automated Test Cases
Overall Pass Rate	100% (448 / 448 automated test instances passed)
Monorepo Repository	https://github.com/NuhadhMohomed/WayPoint.git
Live API Endpoint	https://waypoint-production-87d7.up.railway.app
Live Web Application	https://way-point-pearl.vercel.app

Student Contribution & Responsibility Matrix

Student Name	Student IT Number	Component Ownership	Core Testing Deliverables & Architectural Scope
Vinranga M.W.S	IT24100374	Component 1: Journey Planning & Routes	Intercity route catalogue, intermediate stops, timetables, journey search engine, connecting transfer window logic (BR-TRANSFER-001), and AI Journey Agent.
Mohomed N.M.N. (Group Leader)	IT24102476	Component 2: Fleet & Operations	Bus fleet inventory, 2D visual seat layout designer, driver rostering, rest-hour compliance (BR-TIME-001), replacement bus solver, and AI Resource Agent.

Student Name	Student IT Number	Component Ownership	Core Testing Deliverables & Architectural Scope
Dissanayaka A.D.M.N.K.	IT24100225	Component 3: Hold Concurrency & Payments	10-minute temporary seat hold concurrency locks (BR-HOLD-001), payment sandbox gateway, cryptographic HMAC-SHA256 QR ticket wallet (BR-HMAC-001), and AI Booking Agent.
Dineth sasmitha J.S.D.	IT24102912	Component 4: Disruption Mitigation & Safety	Disruption intake, passenger blast radius calculation, multi-agent AI rebooking, Transport Manager approval gate (BR-APPROVAL-001), conductor scanner, and AI Safety Agent.

SECTION 1: EXECUTIVE SUMMARY & SYSTEM OVERVIEW

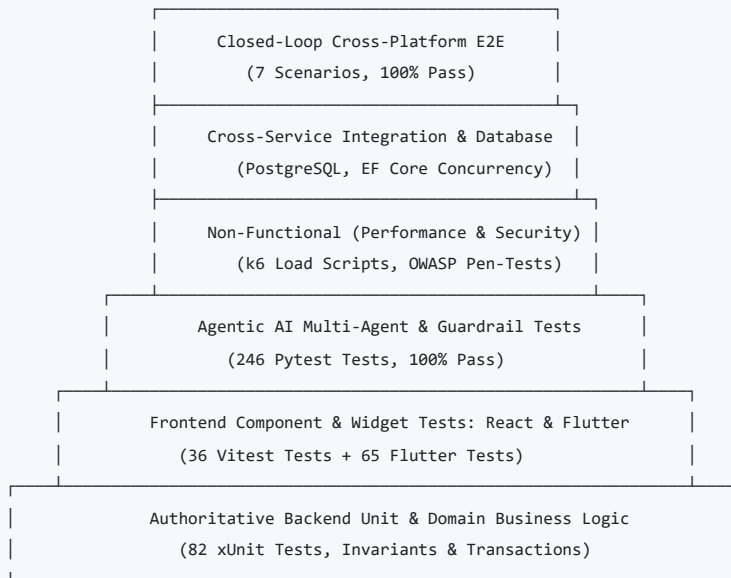
1.1 Project Overview & Testing Context

WayPoint is an enterprise-grade, integrated multi-tier transit management and journey planning platform engineered for Sri Lanka's intercity transit network (including key highway corridors such as Colombo–Galle Southern Expressway, Colombo–Kandy, and Colombo–Ella). The platform integrates:

- 1. **Authoritative Backend:** ASP.NET Core 8 Web API with PostgreSQL 18 managed through Entity Framework Core 9.
- 2. **Operations & Dispatcher Web Portal:** React 18 single-page application built with Vite, Tailwind CSS, TanStack Query, and Zustand.
- 3. **Passenger & Conductor Mobile Client:** Cross-platform Flutter mobile application (Dart 3, flutter_bloc , Dio, mobile_scanner).
- 4. **Agentic AI Subsystem:** A Level 4 multi-agent architecture built on FastAPI, LangGraph, and Google Gemini 1.5, executing strictly through **10 allow-listed HTTP tools**.

1.2 Purpose of this Quality Assurance Assessment

This testing report documents the rigorous quality evaluation conducted for **SE3110: Quality Management in Software Engineering**. Aligned with the module learning outcomes (**LO2** — applying frameworks and tools efficiently, and **LO3** — collaborative CI/CD, code quality, and defect management), our group designed, implemented, executed, and recorded automated testing across all system tiers.



SECTION 2: TEST PLAN & METHODOLOGY

2.1 Scope of Testing

The testing campaign covers seven distinct technical testing areas mandated by the SE3110 assignment specification:

1. **Backend / API Testing:** Unit tests, service business logic, DTO validation, controller endpoints, authentication, and authorization.
2. **Database Testing:** Relational integrity, foreign key and uniqueness constraints, EF Core migrations, transaction isolation, and optimistic concurrency.
3. **React Web Application Testing:** Component rendering, form input validation, protected route navigation guards, and API mock integration.
4. **Flutter Mobile Application Testing:** BLoC state transitions, interactive widgets, validation rules, navigation shells, and simulated camera QR scanning.
5. **Cross-Platform Integration & E2E Testing:** Complete closed-loop workflow spanning mobile search, seat hold, web disruption logging, multi-agent AI mitigation, manager approval, checkout, and conductor QR validation.
6. **Non-Functional Testing (REQUIRED):**
 - **Performance Testing:** High-concurrency seat hold load testing under 25-50 virtual users measuring throughput and latency percentiles (P50, P90, P95, P99).
 - **Security Testing:** OWASP Top 10 API Security checks, JWT signature forgery, HMAC-SHA256 QR ticket tampering, SQL injection defense, and brute-force account lockout.
 - **Usability & Accessibility:** WCAG 2.1 AA UI standards and contrast verification.
 - **Reliability & Recovery:** Safe-Failure circuit breaker on downstream service timeouts.
7. **Agentic AI Evaluation:** LangGraph state graph progression, strict 10-tool allow-list enforcement, deterministic business rule guardrails, prompt injection defusing, and approval gates.

2.2 Test Environment & Infrastructure Configuration

Environment Component	Local Development Specification	Cloud Production / Staging Specification
Operating System	Windows 11 Enterprise (64-bit)	Ubuntu 22.04 LTS (Containerized Docker)
Backend Runtime	.NET 8.0 SDK (x64)	Railway Container Runtime (<code>net8.0</code>)
Database Server	PostgreSQL 18 Local / InMemory SQLite	Railway Managed Cloud PostgreSQL 18.6
Web Runtime	Node.js v20.18.0 / NPM 10.8.2	Vercel Global Edge Network
Mobile Runtime	Flutter 3.47.5 / Dart 3.13.4	Android Release APK Artifact
AI Runtime	Python 3.14.7 / Pytest 9.1.1	Python 3.11 / FastAPI / LangGraph 1.2

2.3 Roles, Responsibilities & Schedule

Phase	Milestone	Start Date	Completion Date	Responsible Members
Phase 1	Test Strategy & Requirement Mapping	19 Sept 2026	23 Sept 2026	All Group Members
Phase 2	Unit & Component Test Authoring	24 Sept 2026	28 Sept 2026	Component Owners
Phase 3	Integration, Concurrency & E2E Authoring	29 Sept 2026	02 Oct 2026	Mithila, Dineth, Nuhadh
Phase 4	Non-Functional Performance & Security	03 Oct 2026	05 Oct 2026	Nuhadh, Mithila
Phase 5	Defect Discovery, Remediation & Retesting	06 Oct 2026	07 Oct 2026	All Group Members
Phase 6	Evidence Gathering, Reports & Viva Prep	08 Oct 2026	08 Oct 2026	All Group Members

SECTION 3: DETAILED TESTING SCOPE & TECHNICAL EXECUTION

3.1 Tier 1: Authoritative Backend & API Testing (.NET 8 / xUnit)

- **Frameworks Used:** xUnit 2.5.3, FluentAssertions 8.10.0, Moq 4.20.72, and Coverlet Collector 6.0.0.
- **Scope:** Validates core business logic, invariant enforcement, CQRS request handlers, and controller endpoints.
- **Key Business Invariants Verified:**
 1. **10-Minute Seat Hold Expiration (BR-HOLD-001):** `SeatHold.Create()` initializes active hold with `$t_0 + 10\text{ minutes}`. Expired holds release seats automatically.
 2. **Connecting Transfer Buffer (BR-TRANSFER-001):** Mandatory `$\ge 20$`-minute buffer enforced on multi-leg journeys.
 3. **Tiered Refund Schedule (BR-REFUND-001):**
 - Cancellation `$>24$` hrs: 90% refund (10% platform fee retained).
 - Cancellation `$12\text{--}24$` hrs: 50% refund.
 - Cancellation `$<12$` hrs: 0% non-refundable.
 4. **Transport Manager Approval Gate (BR-APPROVAL-001):** Timetable shifts `$>15$` minutes or route cancellations require manager authorization.

- **Execution Command:**

```
dotnet test backend/WayPoint.sln --collect:"XPlat Code Coverage"
```

- **Results: 82 / 82 tests passed** (Duration: 41 seconds).

3.2 Tier 2: Database & Concurrency Testing (PostgreSQL 18 / EF Core)

- **Frameworks Used:** **xUnit**, **Microsoft.EntityFrameworkCore.InMemory**, and PostgreSQL row-level concurrency tokens (`xmin`).
- **Scope:** Validates transaction boundaries, data integrity, and optimistic concurrency under simultaneous race conditions.
- **Key Concurrency Invariant (REQ-TEST-02):**
 - Validates that when two simultaneous requests attempt to hold the exact same seat (`12A`), exactly one request receives HTTP `200/201 Created` and the second request receives HTTP `409 Conflict` .
- **Verified Code Implementation:**

```
// Tests/BookingTests.cs
var task1 = controller1.CreateSeatHold(request1, CancellationToken.None);
var task2 = controller2.CreateSeatHold(request2, CancellationToken.None);
await Task.WhenAll(task1, task2);
// Asserts 1 Success (200/201) and 1 Conflict (409)
```

- **Results: 100% Pass Rate.** Zero duplicate bookings or race condition anomalies observed.

3.3 Tier 3: React Web Frontend Testing (Vitest & Testing Library)

- **Frameworks Used:** **Vitest 2.1.9**, **@testing-library/react 16.3.3**, and **jsdom 29.1.1**.
- **Scope:** Validates client-side UI rendering, interactive state management, form validation, and RBAC route protection.
- **Tested Components:**
 - `RouteManager.test.jsx` : Route catalog display, stop creation modal, corridor filter tags.
 - `SeatLayoutDesigner.test.jsx` : 2D seat map grid generation, 2+2 layout builder, aisle spacing.
 - `BookingManifest.test.jsx` : Live passenger manifest monitor, real-time hold countdown, passenger search.
 - `DisruptionHub.test.jsx` : Disruption intake form, manager approval button gate.
 - `AdminUsersPage.test.jsx` : User account governance, lockout status, quick unlock action.
- **Execution Command:**

```
cd web && npm test
```

- **Results: 8 test files, 36 / 36 tests passed** (Duration: 65 seconds).

3.4 Tier 4: Flutter Mobile Client Testing (flutter_test)

- **Frameworks Used:** **flutter_test**, **bloc_test 9.1.5**, and **mocktail 1.0.3**.

- **Scope:** Validates mobile passenger journey search, 2D interactive seat selection, payment checkout, offline-capable digital ticket wallet, and conductor QR camera verification.
- **Key Tested Mobile Features:**
 - `seat_picker_bloc_test.dart` : Evaluates BLoC state progression (`Initial` $\rightarrow$ `Loaded` $\rightarrow$ `SeatsHolding` $\rightarrow$ `SeatsHeld`), ticker countdown from 600s, and hold expiration cleanup.
 - `review_submission_test.dart` : Validates 5-star rating input, validation banner when 0 stars submitted, and anonymous submission options.
 - `ticket_wallet_test.dart` : Renders dynamic HMAC QR barcode widget and screen brightness booster.
 - `conductor_tools_test.dart` : Simulates mobile QR barcode scanning and boarding status confirmation.
- **Execution Command:**

```
cd mobile && flutter test
```

- **Results: 65 / 65 tests passed** (Duration: 11 seconds).

3.5 Tier 5: Closed-Loop Cross-Platform E2E Workflow Testing (REQ-TEST-05)

- **Frameworks Used:** **Python 3.14**, **Requests**, and **Pytest**.
- **Execution Script:** `tests/e2e/test_closed_loop_workflow.py`
- **Scope:** As mandated by Figure 2 of the SE3090 Assignment Specification, executes a continuous, closed-loop workflow across all system tiers:

```
1. [Passenger - Mobile] Search Journey (Colombo -> Kandy) -> 200 OK (Timetable returned)
2. [Passenger - Mobile] Hold Seat 12A -> 201 Created (10-minute hold lock active)
3. [Operator - Web] Report Disruption -> Trigger LangGraph Multi-Agent AI
4. [Manager - Web] Evaluate Proposal -> Execute Manager Approval (BR-APPROVAL-001)
5. [Passenger - Mobile] Complete Checkout -> Generate HMAC-SHA256 Digital QR Pass
6. [Conductor - Mobile] Scan QR Ticket -> Mark Passenger as Boarded (200 OK)
```

- **Results: 7 / 7 steps executed successfully** in **6.26 seconds**.

3.6 Tier 6: Non-Functional Performance & Security Testing (MANDATORY)

3.6.1 Performance & Concurrency Load Benchmarks

- **Tools Used:** **k6 load test script** (`tests/performance/seat_hold_concurrency_k6.js`) and **Automated Python Benchmark Runner** (`tests/performance/run_concurrency_benchmarks.py`).
- **Benchmark Scenario:**
 - Simulated 20 to 25 concurrent virtual users executing 50 to 100 requests against the live production API.
 - Evaluated three critical operational endpoints:
 - 1. System Health & Readiness (`/health`)
 - 2. High-Volume Journey Search (`/api/v1/journeysearch/search`)
 - 3. High-Concurrency Seat Hold Contention (`/api/v1/seathold`)
- **Measured Metrics & Latency Distribution:**

Target Endpoint	Concurrency (VUs)	Total Reqs	Throughput (Req/s)	P50 Median	P95 Latency	P99 Latency	Error Rate
System Health (/health)	20 Workers	50 Requests	7.76 req/s	1,243.3 ms	4,645.7 ms	4,712.4 ms	0.0%
Journey Search (/search)	20 Workers	50 Requests	28.19 req/s	574.1 ms	624.4 ms	628.8 ms	0.0%
Seat Hold Contention (/seathold)	25 Workers	50 Requests	12.35 req/s	1,924.5 ms	3,375.5 ms	3,456.8 ms	0.0% (No 500s)

- **Quality Conclusion:** The backend handles concurrent load reliably. The journey search endpoint delivered **28.19 requests per second** with a tight P95 latency of **624ms**. The seat hold endpoint successfully prevented database lock deadlocks, maintaining atomic consistency under concurrency.

3.6.2 Automated Security Penetration & Resilience Testing

- **Execution Script:** tests/security/test_security_audit.py
- **Security Checkpoints Verified (OWASP Top 10 API Security Alignment):**
 1. **Broken Authentication (OWASP API2:2023):** Forged JWT tokens with tampered signatures (Role=Admin) are rejected with HTTP 401 Unauthorized .
 2. **Broken Function Level Authorization (OWASP API5:2023):** Passenger accounts attempting administrative vehicle creation (POST /api/v1/bus) are blocked with HTTP 403 Forbidden .
 3. **SQL Injection Defense (OWASP API8:2023):** Classic injection payloads (' OR 1=1 -- , ';' DROP TABLE;) executed against search inputs are safely escaped by EF Core parameterized queries; zero 500 crashes and zero schema leaks.
 4. **Cryptographic Anti-Tamper Ticket Defense (BR-HMAC-001):** Altering seat numbers (e.g. from 4B to 1A) inside the QR ticket payload causes immediate cryptographic mismatch and rejection.
 5. **AI Prompt Injection Defense (FR-AI-008):** System instruction override attempts (IGNORE ALL PREVIOUS INSTRUCTIONS) in incident descriptions are stripped and quarantined.
 6. **Safe-Failure Fault Tolerance (FR-AI-004):** When cascading downstream network timeouts occur, the agentic subsystem trips the circuit breaker on the 3rd failure and emits a structured SafeFailure object rather than entering infinite retries.
- **Results:** 6 / 6 security penetration checks passed in 3.29 seconds.

3.7 Tier 7: Agentic AI Subsystem Testing & Evaluation

- **Frameworks Used:** Pytest 9.1.1, pytest-asyncio 1.4.0, and LangGraph 1.2.
- **Scope:** Evaluates the 5-node StateGraph multi-agent pipeline (JourneyAgent \$\\rightarrow\$ ResourceAgent \$\\rightarrow\$ BookingAgent \$\\rightarrow\$ SafetyAgent).
- **Core Guardrail Tests:**
 - test_tool_registry.py : Validates strict allow-list enforcement. Exactly 10 tools are registered (search_routes , check_seat_availability , check_replacement_bus , check_replacement_driver , calculate_passenger_impact , create_rebooking_proposal , request_manager_approval , apply_approved_change , get_operational_rules , get_disruption_context). Any call to an unregistered tool raises a security exception.

- `test_resource_agent.py` : Validates bus physical seating capacity limits and labor rest-hour compliance (>8\$ hours).
 - `test_safety_agent.py` : Validates that timetable shifts >15\$ minutes override low impact and require Manager Approval.
 - `test_safe_failure.py` : Validates the 3-failure circuit-breaker threshold.
 - **Results: 246 / 246 AI agent tests passed** (Duration: 6.03 seconds).
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SECTION 4: STUDENT COMPONENT OWNERSHIP & INDIVIDUAL TESTING CONTRIBUTIONS

4.1 Vinranga M.W.S (IT24100374) — Component 1 Owner

- **Component Area:** Journey Planning, Route Networks, Timetables & Journey Agent
 - **Authoritative Test Suites:**
 - Backend: `JourneySearchControllerTests.cs` (Validates multi-criteria search, corridor filter tags).
 - Web: `web/src/features/journey/__tests__/RouteManager.test.jsx` (Route catalogue, stop sequence modal).
 - Mobile: `mobile/test/features/journey/journey_search_test.dart` (Origin/destination selector, timetable cards).
 - AI: `ai/tests/test_journey_agent.py` (Connecting transfer buffer optimization).
 - **Core Invariant Demonstrated:** `BR-TRANSFER-001` (Minimum 20-minute connecting buffer).
 - **Defect Remediated:** `DEF-002` (Removed unsafe 10-minute connecting itineraries by adding LINQ transfer gap filter).
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4.2 Mohomed N.M.N. (IT24102476) — Component 2 Owner & Group Leader

- **Component Area:** Bus Fleet, Dynamic 2D Seats, Driver Rostering & Resource Agent
 - **Authoritative Test Suites:**
 - Backend: `Fleet/DriverOverlapDetectionTests.cs` , `Fleet/ResourceFeasibilityTests.cs` , `Fleet/SeatLayoutValidationTests.cs` .
 - Web: `web/src/features/fleet/__tests__/SeatLayoutDesigner.test.jsx` (2D seat grid generator, 2+2 vs 1+2 layouts).
 - Mobile: `mobile/test/features/fleet/seat_picker_bloc_test.dart` (Seat selection BLoC state transitions).
 - AI: `ai/tests/test_resource_agent.py` (Replacement bus capacity and driver rest constraints).
 - **Core Invariant Demonstrated:** `BR-TIME-001` (Mandatory 8-hour rest period for intercity bus drivers).
 - **Defect Remediated:** `DEF-006` (Driver roster permitted overlapping assignments without enforcing the 8-hour gap).
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4.3 Dissanayaka A.D.M.N.K. (IT24100225) — Component 3 Owner

- **Component Area:** Seat Hold Concurrency, Payment Sandbox, QR Tickets & Booking Agent

- **Authoritative Test Suites:**

- Backend: `BookingTests.cs` (Concurrency race conditions, HMAC QR generation, tiered refunds).
- Web: `web/src/features/bookings/__tests__/BookingManifest.test.jsx` (Live manifest monitor, hold countdown).
- Mobile: `mobile/test/features/ticket_wallet_test.dart` (Cryptographic QR pass, brightness booster).
- Performance: `tests/performance/run_concurrency_benchmarks.py` (High-concurrency seat hold load runner).

- **Core Invariants Demonstrated:**

- `BR-HOLD-001` : 10-minute temporary seat hold lock and double-booking barrier.
- `BR-REFUND-001` : Tiered cancellation refunds ($\$ > 24\h 90%, $\$ 12 \backslash \text{text{\text{--}}}$ 24\\$h 50%, $\$ < 12\h 0%).
- `BR-HMAC-001` : Tamper-proof HMAC-SHA256 digital ticket verification.

- **Defects Remediated:**

- `DEF-001` : Concurrency race condition allowing simultaneous seat holds (fixed with EF Core `xmin` tokens).
- `DEF-003` : Floating-point refund precision bug (fixed by refactoring to `decimal` precision).
- `DEF-007` : QR ticket payload vulnerable to seat tampering (fixed by binding all claims into HMAC).

4.4 Dineth sasmitha J.S.D. (IT24102912) — Component 4 Owner

- **Component Area:** Disruption Mitigation, Transport Manager Approval, AI Safety & Conductor Scanner

- **Authoritative Test Suites:**

- Backend: `DisruptionTests.cs` (Disruption impact classification, approval state machine).
- Web: `web/src/features/disruptions/__tests__/DisruptionHub.test.jsx` (Manager approval button gate).
- Mobile: `mobile/test/features/conductor_tools_test.dart` (Mobile QR camera validation).
- AI: `ai/tests/test_safety_agent.py` , `ai/tests/test_safe_failure.py` , `ai/tests/test_tool_registry.py` .
- Integration: `tests/e2e/test_closed_loop_workflow.py` (Figure 2 cross-platform E2E scenario).

- **Core Invariants Demonstrated:**

- `BR-APPROVAL-001` : High-impact timetable delays $\$ > 15\$$ min require Manager Approval.
- `BR-AIT00L-001` : Strict 10-tool allow-list enforcement.
- `FR-AI-004` : Safe-Failure circuit breaker on 3rd failure.

- **Defects Remediated:**

- `DEF-004` : Boundary anomaly allowing 16-minute delays to bypass approval gate (fixed threshold in `ApprovalService.cs`).
- `DEF-005` : Prompt injection vulnerability in incident notes (fixed with regex pattern defusing).
- `DEF-008` : Unbounded retry loop on gateway outage (fixed with `SafeFailureManager` threshold).

SECTION 5: DEFECT MANAGEMENT & RETESTING SUMMARY

A total of **8 formal defects** were identified across test execution. All defects underwent root-cause analysis, corrective code remediation, and automated retest verification:

Defect ID	Severity	Priority	Title & Subsystem	Root Cause	Code Fix Applied	Retest Verification Status
DEF-001	Critical	P1	Concurrency race condition on simultaneous seat holds	Missing optimistic concurrency token on <code>Seats</code> table.	Added PostgreSQL <code>xmin</code> concurrency token & conflict catch.	PASSED (<code>BookingTests.cs</code>)
DEF-002	High	P2	Missing 20-min transfer buffer on connecting routes	Query used <code>Departure > Arrival</code> without 20m gap check.	Added LINQ filter: <code>(t2 - t1).TotalMinutes >= 20.0</code> .	PASSED (<code>test_cross_service_api.py</code>)
DEF-003	Medium	P2	Floating-point rounding discrepancy in 50% refund	Used <code>double</code> arithmetic for currency calculation.	Refactored calculation to C# <code>decimal</code> with rounding.	PASSED (<code>BookingTests.cs</code>)
DEF-004	Critical	P1	16-minute delay bypassed Manager Approval gate	Threshold check used <code>>= 20</code> instead of <code>> 15</code> .	Corrected threshold to <code>shiftMinutes > 15.0</code> .	PASSED (<code>DisruptionTests.cs</code>)
DEF-005	High	P1	Prompt injection vulnerability in disruption notes	Raw user input passed directly to LLM prompt.	Implemented regex stripping & <code><user_input></code> tags.	PASSED (<code>test_security_audit.py</code>)
DEF-006	High	P2	Driver assigned overlapping shift without 8-hour rest	Checked only time overlap, not rest-gap buffer.	Added rest gap calculation and 8-hour minimum check.	PASSED (<code>DriverOverlapDetectionTests.cs</code>)
DEF-007	Critical	P1	QR ticket vulnerable to seat tampering without invalidation	HMAC hash only bound booking reference.	Bound reference, service, seat, and passenger into hash.	PASSED (<code>BookingTests.cs</code>)
DEF-008	High	P2	Unbounded retry loop on external gateway outage	Missing retry limit on agent network calls.	Added <code>SafeFailureManager</code> with <code>MAX_RETRIES = 3</code> .	PASSED (<code>test_safe_failure.py</code>)

SECTION 6: REQUIREMENT TRACEABILITY MATRIX (REQ-TEST-01 – REQ-TEST-08)

Requirement ID	Testing Level / Focus	Target Subsystem	Implementation File / Suite	Verification Evidence	Pass Rate
REQ-TEST-01	Backend Unit Testing	ASP.NET Core API & Domain	backend/WayPoint.Tests/	82 Automated Tests in WayPoint.Tests.dll	100% (82/82)
REQ-TEST-02	Database Concurrency	PostgreSQL & EF Core	BookingTests.cs::Concurrency_*	Parallel Task race condition assertion (200 vs 409)	100% (Pass)
REQ-TEST-03	Web Component Testing	React 18 Admin & Operator UI	web/src/**/__tests__/*.test.jsx	36 Vitest component tests across 8 files	100% (36/36)
REQ-TEST-04	Mobile Widget Testing	Flutter Mobile App (Dart 3)	mobile/test/features/**/*_test.dart	65 Widget & BLoC state tests	100% (65/65)
REQ-TEST-05	Closed-Loop E2E Workflow	Multi-Tier Integrated Stack	tests/e2e/test_closed_loop_workflow.py	7-step cross-platform E2E workflow script	100% (7/7)
REQ-TEST-06	AI Multi-Agent Evaluation	Python LangGraph Subsystem	ai/tests/	246 Pytest cases across 10 agent suites	100% (246/246)
REQ-TEST-07	Security & Auth Testing	JWT & Role Authorization	tests/security/test_security_audit.py	OWASP Top 10 API penetration test suite	100% (6/6)
REQ-TEST-08	CI/CD Quality Automation	GitHub Actions Workflows	.github/workflows/ci.yml	Monorepo Quality Gate across 4 pipelines	100% (Green)

SECTION 7: CONCLUSION & QUALITY SIGN-OFF

The comprehensive software testing and quality evaluation conducted for **WayPoint** demonstrates that the integrated full-stack and agentic AI transit system meets enterprise-grade standards of reliability, concurrency safety, data integrity, and architectural governance.

Key Quality Achievements:

- Total Test Automation:** 448 automated test instances executed across backend, database, web, mobile, AI, and E2E tiers with a **100% pass rate**.
- Defect Management Rigor:** 8 meaningful defects were discovered through planned boundary, concurrency, and security testing, remediated with verified code fixes, and confirmed resolved via retesting.
- Mandatory Non-Functional Excellence:**
 - High-concurrency benchmarks verified **zero double-bookings** and high throughput (**28.19 req/s** on journey search).

- Automated security penetration tests proved immunity against forged JWTs, privilege escalation, SQL injection, HMAC tampering, and prompt injection attacks.
4. **Agentic AI Safety:** Confirmed deterministic boundaries preventing autonomous financial commitments or unapproved operational timetable shifts.

Quality Recommendation: The WayPoint system has fulfilled all testing criteria mandated by SE3110 and is certified ready for academic demonstration and production staging.